

SESSION - 04

Developer Tools



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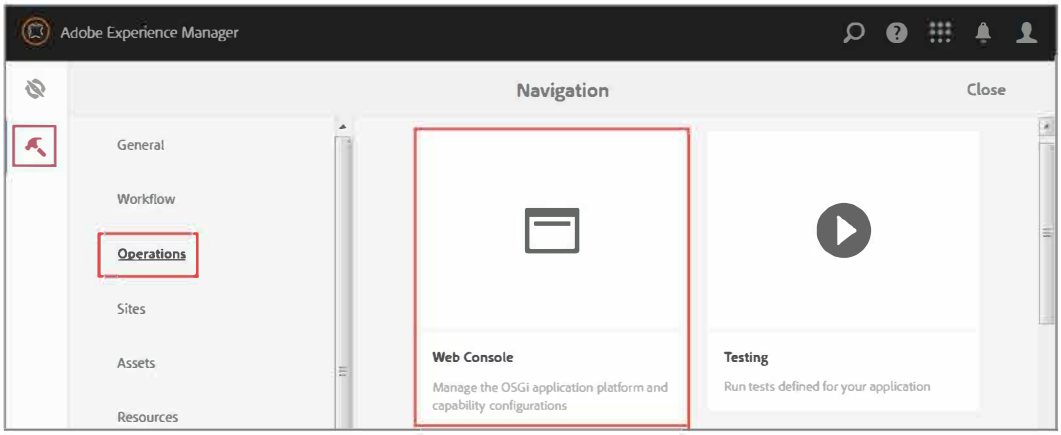
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DEVELOPER TOOLS

The Web Console

The Web Console in Adobe Experience Manager is based on the Apache Felix Web Management Console. It is used to manage various bundles and configurations. Any changes made through this console are automatically applied to the running system, without the need to restart the instance.

You can access the console at: http://localhost:4502/system/console or by navigating to Tools > Operations > Web Console, as shown:



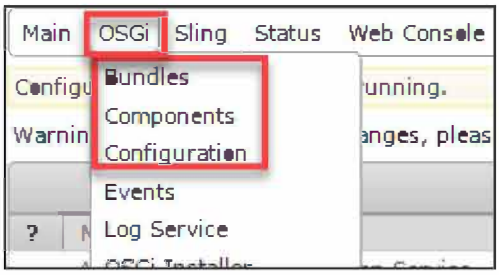
If you use the navigation in AEM, the Web Console opens in a new tab in your browser.

The most important menu selections under the OSGi tab are:

Bundles—used for installing and managing bundles.

Components—used for managing and controlling the status of components required for Adobe Experience Manager.

Configuration—used for configuring OSGi bundles, and is the underlying mechanism for configuring Adobe Experience Manager parameters.

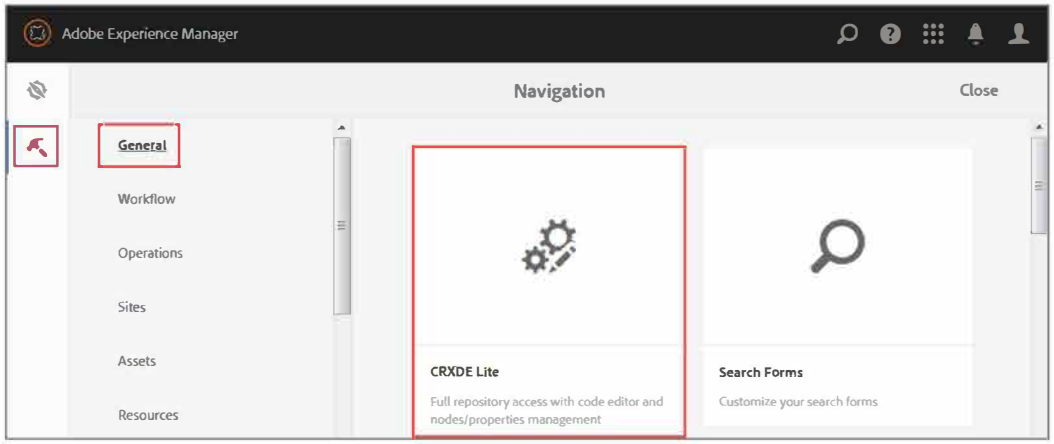


CRXDE Lite

The CRXDE Lite console is embedded into Adobe Experience Manager and allows you to perform common development and administration tasks in the browser. Because it is embedded in the server and is always available, CRXDE Lite is often the preferred tool for administrators and developers for working with nodes and properties in the JCR. It gives quick and direct access to the repository for monitoring, configuration, and development.

NOTE: You will use this interface frequently in this training.

You can access this console through http://localhost:4502/crx/de/index.jsp or by navigating to Tools > General > CRXDE Lite from the Touch UI, as shown:



With CRXDE Lite, you can create a project, create and edit files (like .jsp and .java), folders, templates, components, dialogs, nodes, properties, and bundles. CRXDE Lite is recommended for most repository-level administration tasks as well as many light weight development tasks.

The CRXDE Lite console contains:

Top switcher bar: Enables you to quickly switch between CRXDE Lite, Package Manager, and Package Share.

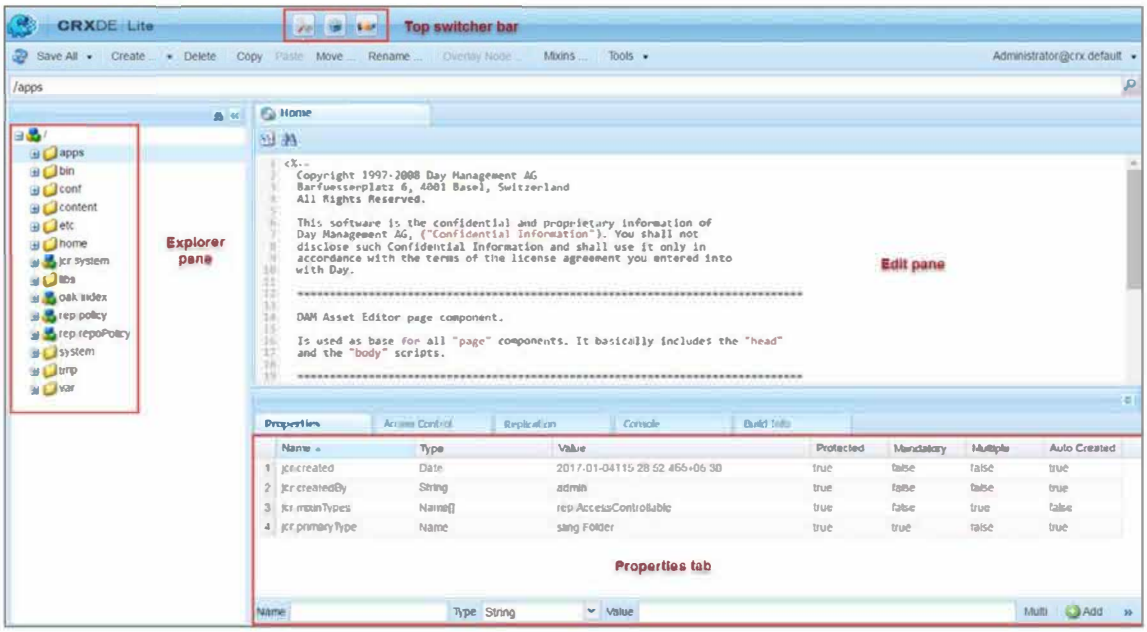
Explorer pane: Displays a tree of all the nodes in the repository.

You can perform the following actions on a node in the tree:

- Select the node and view its properties in the Properties tab. Examine all the JCR properties of different nodes.
- Right-click the node and perform an action on it, such as renaming the node, creating a new node, creating a folder, creating a file.

Edit pane: Double-click a file in the Explorer pane to display its content. For example, a .jsp or a .html file. You can then modify the code and save the changes.

Properties tab: Displays the properties of the node that you selected. You can add new properties or delete existing ones.



Working with Packages

A package is a zip file that holds repository content in the form of a file-system serialization called "vault" serialization. Packages provide an easy-to-use-and-edit representation of files such as pages, assets, and folders.

Throughout this training, you will be working with packages; either by creating your own or installing an available package. These packages enable you to import and export repository content. A package may contain:

- Page-related content
- Project-related content
- Assets-related content
- Vault meta information such as filter definitions and import configuration information
- Other package information such as package settings, package filters, package screenshots, and package icons

Packages enable you to import and export repository content.

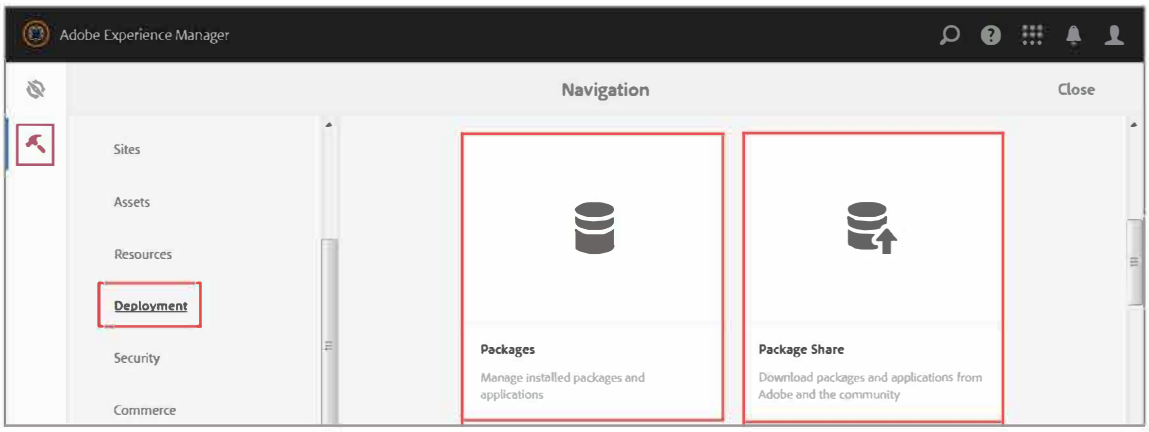
You commonly use packages for any of the following:

- Install new functionality
- Transfer content between instances
- Back-up repository content
- Export content to the local file system

You can work with packages in two ways:

1. **Package Manager**—can be used to import or export content, transfer content between instances, and back-up repository content. Using filters, you can create a package to contain page content, or project-related content. You can access this console through the following link: http://localhost:4502/crx/packmgr/index.jsp. With the Package Manager, you can perform the following common tasks:
    - Create, build, and download content packages
    - Upload and install packages
    - Modify existing packages
    - View package information
2. **Package Share**—a centralized server that contains both public and private packages available across all instances. Public packages may include hotfixes, new functionality, documentation, and so on. You can access this console through the following link: http://localhost:4502/crx/packageshare/index.html.

    To access Package Manager and Package Share from the Touch UI, navigate to **Tools > Deployment > Packages/ Package Share** as shown:



Creating and Building New Packages

When creating packages using Package Manager, you can apply rules and filters to determine the content package should extract from the repository. After you define the content, you can build it. A package is created in a .zip file and you can download it to your local file system. You can test the contents of the package before building it.
There are many options in Package Manager to work with packages, such as:

- Rebuild: Helps rebuild the package if there is a change in the repository content.
- Edit: Helps edit filters or rules applied to the package.
- Test: Helps perform a dry run of the installation.
- Rewrap: Helps recreate the package with additional information such as thumbnails and icons.

Downloading packages to your file system

You can download a package by clicking the download link. This link displays when the package details are expanded. After downloading the package, you can unzip the contents of the package onto your local system.



Typically, a content package contains the following folders:

- **jcr_root:** Represents the root node of the repository and contains the actual content of the package.
- **META-INF**: Contains metadata regarding node definitions and the filter.xml file that gives directions to FileVault about the paths to include.

Using the Package Share

Package Share is a centralized server where public and private packages are made available. These packages may be hotfixes, feature sets, updates, or Adobe Experience Manager content generated by other users. You can search, download, and install any package either to your instance or to your local file system.

Within the Package Share, you have access to the following:

- Adobe packages provided by Adobe
- Shared packages provided by others companies and made public by Adobe

  Your company packages that are private

  You can access the Package Share with this link: http://localhost:4502/crx/packageshare/login.html
- You can directly access Package Share (without using CRXDE Lite) through the following link: https://www.adobeaemcloud.com/content/packageshare.html

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NOTE: You must have an Adobe ID to use the Package Share. For more information, refer to the online documentation about Package Share.

Exercise

Task - Install a package in AEM

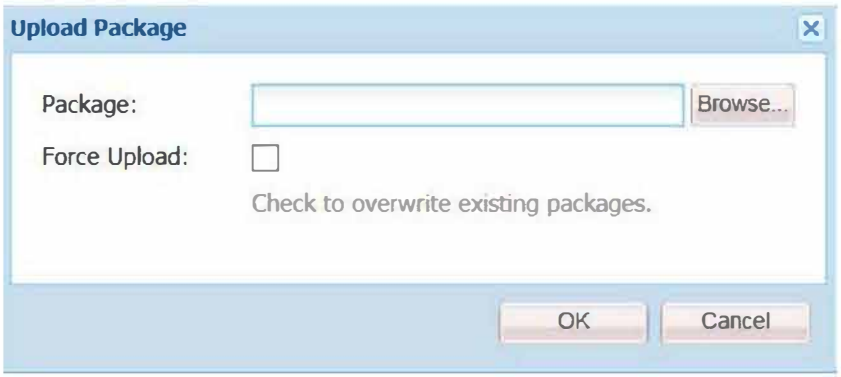
1. Navigate to the Projects console, or click the following link: http://localhost:4502
2. Click Adobe Experience Manager in the upper-left, then click the Tools icon.



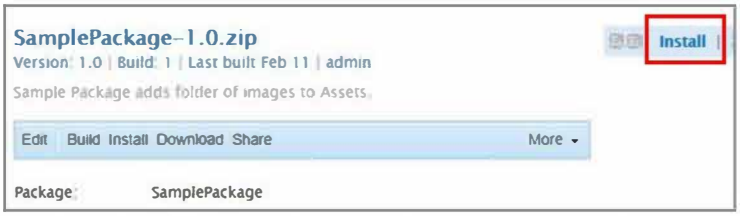
3. Under the Deployment section, select Packages. This opens the Package Manager of CRXDE Lite. As an alternative, click the following link to open the page directly: http://localhost:4502/crx/packmgr/index.jsp
4. Click Upload Package.



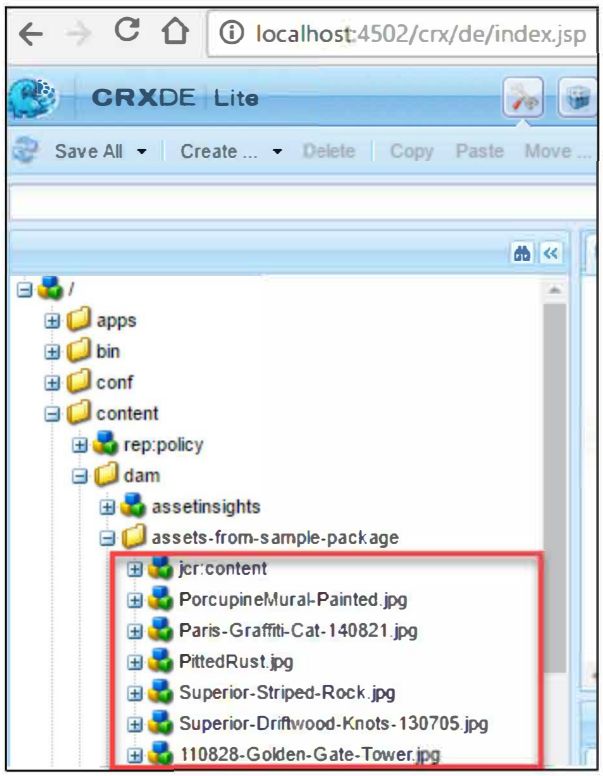
5. In the Upload Package dialog box, click Browse, and select the SamplePackage.zip package from the Exercise_Files provided to you. Click OK.



6. After the package is uploaded, click Install.



7. In the Install Package dialog box, ignore the Advanced Settings area and click Install.
8. Check the Activity Log. You can see the content was added from the package.
9. Check the /content/dam/assets-from-sample-package folder in CRXDE Lite to see the changes reflected there. The contents of the package are added to the repository.


localhost:4502/crx/de/index.jsp
CRXDE Lite
Save All
Create ...
Delete
Copy
Paste
Move ...
/
apps
bin
conf
content
rep:policy
dam
assetinsights
assets-from-sample-package
jcr:content
PorcupineMural-Painted.jpg
Paris-Graffiti-Cat-140821.jpg
PittedRust.jpg
Superior-Striped-Rock.jpg
Superior-Driftwood-Knots-130705.jpg
110828-Golden-Gate-Tower.jpg

Task - Create, Build, and Download a Package

1. **Click the Package Manager button in CRXDE Lite. This takes you back to the Package Manager.**

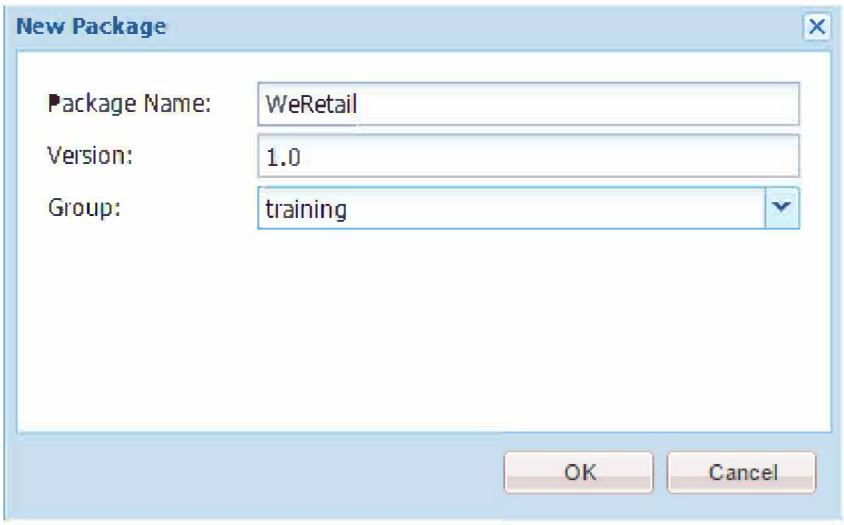


2. **As an alternative, click the following link to open the page directly: http://localhost:4502/crx/packmgr/index.jsp**
3. **Click Create Package.**

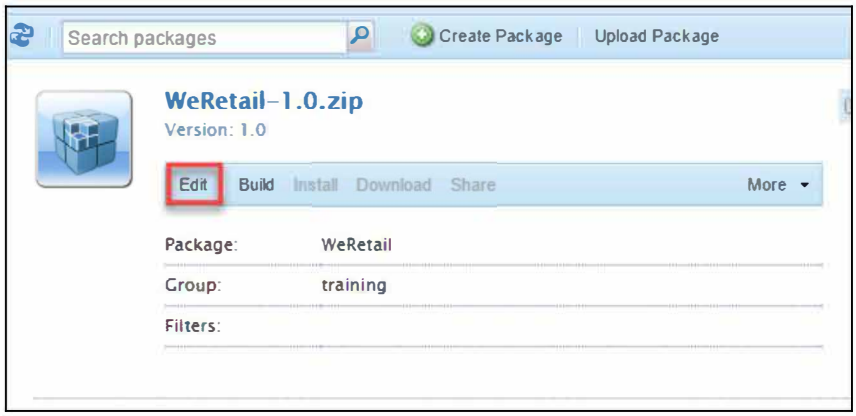


4. **In the New Package dialog box, enter the following details:**

Property	Value
Package Name	WeRetail
Version	1.0
Group	training

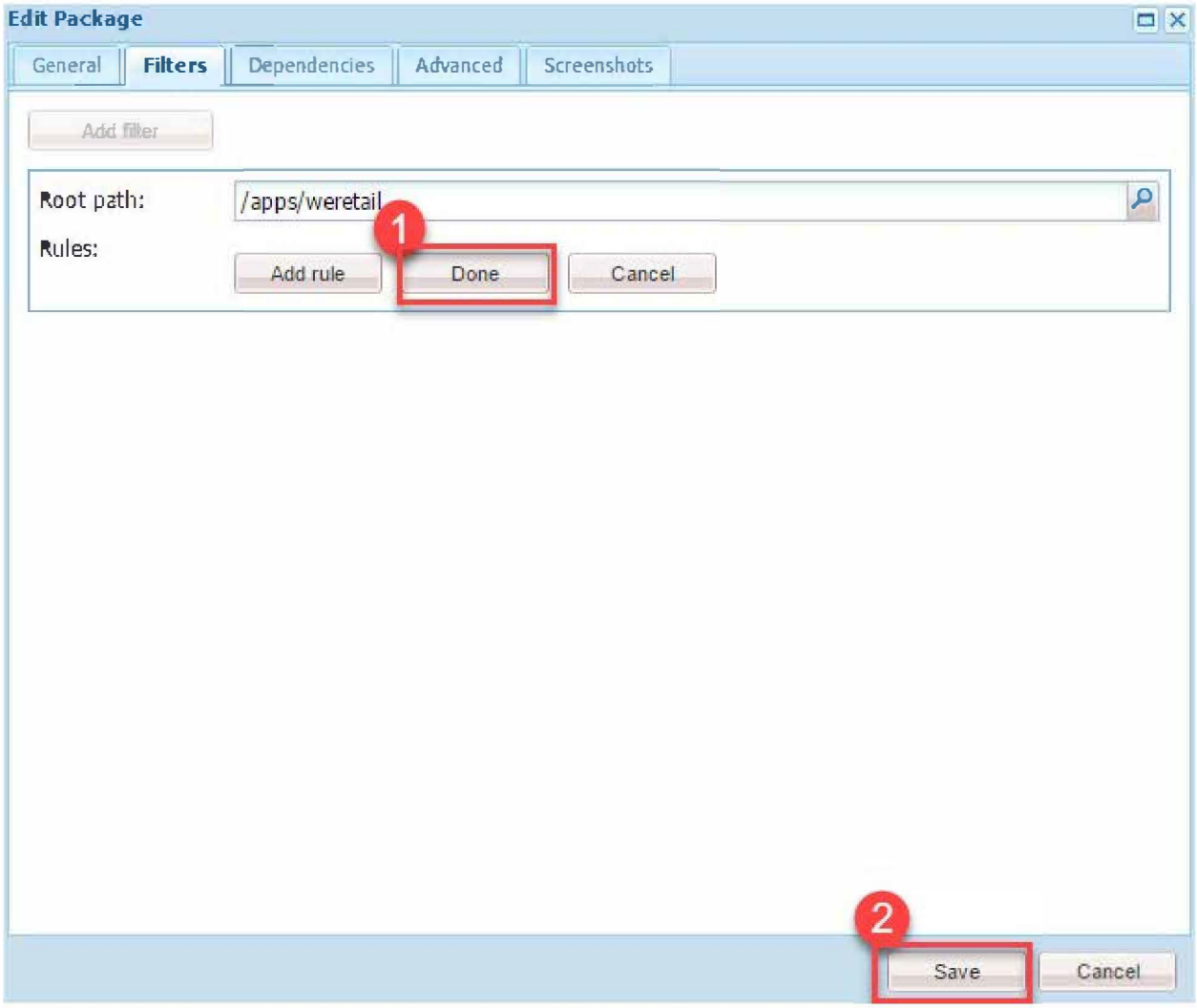


5. **Click OK.**
6. **Click Edit on the newly-created package.**



7. **To add filters to the package, click the Filters tab, and click Add Filter.**

8. **For the Root path, browse and select the WeRetail project folder under apps, and click OK.**
9. **Click Done, and then click Save.**



10. **Click Build to build the package, and then click Build again in the confirmation dialog box.**



11. **The package is now ready for download. Click the Download link to download a copy of the package to your computer.**